

MMD V2 - Database Design

DATABASE OVERVIEW

Database Engine: PostgreSQL

ORM: Prisma

Architecture: Multi Tenant SaaS

Pattern: Shared Database Shared Schema Tenant Isolation

Every business table contains: tenantId

This ensures tenant-level isolation.

DATABASE STANDARDS

Every table contains: id createdAt updatedAt createdBy updatedBy

Optional: deletedAt

Soft delete strategy: deletedAt IS NULL

CORE IDENTITY MODULE

tenants Purpose: Represents customer organizations. Fields: id name slug logo primaryColor domain status planId createdAt updatedAt Relationships: Tenant → Users → Companies → Leads → Requirements → Candidates → Placements

plans Purpose: Subscription plans. Fields: id name maxUsers maxStorage maxCandidates features

users Purpose: Platform users. Fields: id tenantId roleId name email phone passwordHash status lastLoginAt Relationships: User → Role → Activities → Audit Logs Indexes: email UNIQUE tenantId + email UNIQUE

roles Purpose: Role management. Fields: id tenantId name description Examples: SUPER_ADMIN OWNER ADMIN MANAGER RECRUITER SCRAPER VIEWER

permissions Fields: id name module action Example: companies.create companies.read companies.update

role_permissions Purpose: Role mapping. Fields: roleId permissionId

CRM MODULE

companies Fields: id tenantId name industry website address status Relationships: Company → Contacts → Leads → Requirements → Placements Indexes: tenantId name status

contacts Fields: id tenantId companyId name email phone designation Relationships: Contact → Company

leads Fields: id tenantId companyId contactId assignedTo source status notes Status: NEW CONTACTED QUALIFIED PROPOSAL_SENT CONVERTED LOST Relationships: Lead → Company → Contact → User

ATS MODULE

requirements Fields: id tenantId companyId title skills experience budget location openings status Status: OPEN IN_PROGRESS ON_HOLD CLOSED Relationships: Requirement → Company → Applications

candidates Fields: id tenantId firstName lastName email phone experience currentCompany currentCTC expectedCTC Relationships: Candidate → Documents → Applications → Interviews Indexes: email phone

candidate_documents Fields: id tenantId candidateId fileName fileUrl version Types: RESUME OFFER OTHER

applications Fields: id tenantId candidateId requirementId status assignedRecruiter Pipeline: SOURCED SCREENED SUBMITTED INTERVIEW OFFERED JOINED REJECTED

interviews Fields: id tenantId candidateId applicationId interviewer scheduledAt status feedback Status: SCHEDULED COMPLETED RESCHEDULED CANCELLED

OPERATIONS MODULE

placements Fields: id tenantId candidateId requirementId joiningDate billingAmount Relationships: Placement → Invoice

timesheets Fields: id tenantId placementId employeeId hoursWorked weekStart weekEnd status Status: SUBMITTED APPROVED REJECTED

invoices Fields: id tenantId placementId invoiceNumber amount dueDate status Status: DRAFT SENT PAID OVERDUE

AUTOMATION MODULE

templates Fields: id tenantId name type subject content Types: EMAIL NOTIFICATION WORKFLOW

workflows Fields: id tenantId name trigger conditions actions status

workflow_runs Fields: id workflowId status startedAt completedAt logs

notifications Fields: id tenantId userId title message type readAt

jobs Purpose: Queue system. Fields: id tenantId type payload status scheduledAt processedAt Status: PENDING PROCESSING COMPLETED FAILED No Redis required initially. Database queue only.

REPORTING MODULE

reports Fields: id tenantId name type filters generatedBy generatedAt

analytics_events Fields: id tenantId eventType entityType entityId metadata createdAt Used for: Dashboards Analytics Future AI

ADMINISTRATION MODULE

audit_logs Fields: id tenantId userId action entity entityId oldValue newValue createdAt

api_keys Fields: id tenantId name keyHash status

webhooks Fields: id tenantId name url events status

webhook_deliveries Fields: id webhookId status requestPayload responsePayload attemptCount

WHITE LABEL MODULE

tenant_branding Fields: id tenantId logo primaryColor secondaryColor domain

tenant_settings Fields: id tenantId timezone currency dateFormat language

INDEX STRATEGY

Always Index: tenantId status createdAt Foreign Keys

Additional Indexes: users(email) companies(name) candidates(email) candidates(phone) requirements(status) placements(joiningDate)

SOFT DELETE STRATEGY

Do not delete business records. Use: deletedAt deletedBy Restore capability required.

AUDIT STRATEGY

Track: Create Update Delete Login Logout Permission Changes Role Changes Store in: audit_logs

FUTURE TABLES

subscription_usage billing_transactions client_portal_users candidate_portal_users resume_parsing_results
ai_recommendations conversation_history knowledge_base

DATABASE GROWTH STRATEGY

Phase 1 Single PostgreSQL Instance ↓ Phase 2 Read Replicas ↓ Phase 3 Partition Large Tables ↓ Phase 4 Data Warehouse Analytics Database

This design supports: 10 Tenants 100 Tenants 1000 Tenants 10000 Tenants without redesigning the database.